



Fluid flow and bubble evolution in the melt pool of high-power (>2500 W) laser additive manufacturing of aluminum captured by X-ray imaging

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ABSTRACT

With the continuous advancement of industrial technology, the demand for large and high-performance parts in industries such as aerospace, automobile, and energy are growing. High-power additive manufacturing can achieve a deep melt pool and fast forming speed, which is particularly important for the manufacturing of mass production, large-size parts and high melting point metals. In view of the fact that the keyhole mode under high-power will induce the pore generation, the in-situ X-ray imaging technology is used in this work to observe the entire process of fluid flow, bubble generation and evolution in the high-power laser melting of aluminum alloy. By comparing and analyzing the metallurgical behaviors of the fluid in the melt pool before and after the laser is turned off, the effect of keyhole on the fluid flow is revealed, that is, the keyhole leads to two relatively independent flow loops in the melt pool. Four bubble formation mechanisms are elucidated, namely, the bubbles formed by entrainment of the shielding gas through the free surface, bubbles formed at the melt pool boundary, bubbles formed at the crack, and bubbles formed at the keyhole bottom. Four bubble evolution mechanisms are also clarified. The bubbles may escape from the free surface, merge into large bubbles, be captured by the keyhole, and be captured by the solidification interface to form pores. We also reveal the dynamic process of bubbles being captured by the solidification interface to form pores. Under the interface constraint, the large-sized bubble splits into a pore and a new small bubble. While the small-sized bubble completely forms a pore. Finally, the characteristic of the bubble in the melt pool after the laser is turned off under high-power printing is clarified, that is, a large number of small bubbles appear and finally turn into pores. This work shows the phenomena in high-power additive manufacturing, which are different from those in low-power additive manufacturing. Two independent circulation flows occur in the obtained large-sized melt pool, and the large-sized bubbles split into small-sized pores during the capture process, as well as a large number of small bubbles occurs in the melt pool after the laser is turned off. The results provide theoretical support for optimizing processes, reducing defects, and improving component performance for high-power additive manufacturing, as well as the welding process.

1. Introduction

The laser powder bed fusion (LPBF), one of the additive manufacturing technologies, using a focused laser beam to scan the

powder bed along the predesigned path and then accumulates layer by layer to obtain the three-dimensional parts, has the advantages of short cycle, high degree of freedom, and integrated forming [1,2]. During the laser forming process, in order to ensure the stability of the melt pool

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and the forming quality, the laser power is generally controlled in the range of 100–1000 W for LPBF technology [3–5]. With the continuous advancement of industrial technology, the demands for large and high-performance parts in industries such as aerospace, automobiles and energy are gradually increasing. High-power additive manufacturing can provide stronger energy input, achieve a deeper melt pool and faster forming speed, thereby significantly shortening the manufacturing time. This is particularly critical for manufacturing of mass production, large-size part and high-melting point metal in the industrial field, and can meet the needs of high-yield and efficient manufacturing. In addition, high-power additive manufacturing can improve the energy utilization through the deep melting mode (keyhole mode), which is particularly suitable for high-reflectivity materials [6]. For example, the absorption rate of aluminum alloy and steel to 1060 nm infrared light is below 20 % and below 35 % in the conduction mode [7], respectively. Studies have shown that the laser absorption rate in keyhole mode can be up to 90 % [8], which is crucial for improving energy efficiency and avoiding damage to the optical path system of the equipment by reflected laser beams.

For LPBF and laser welding processes, which are both based on the high-intensity laser, the basic physical process of high-power additive manufacturing is very similar to the welding process due to the directional and energy distribution characteristics of the laser beam (locally limited energy input). The high-power laser beam causes the temperature of the material in the focal area to quickly reach and exceed the boiling point, generating a large amount of metal vapor. The recoil pressure overcomes the surface tension and gravity, pushing the molten metal to the surroundings to form a keyhole. The keyhole is not a static cavity, but a highly dynamic structure [9]. Its front wall continues to advance due to the continuous ablation of the forward-moving laser, and the rear wall closes due to the backfilling of molten metal. The stability of the keyhole is critical to the laser machining process [10]. An unstable keyhole will expand, contract, or even collapse periodically, trapping bubbles in the melt pool and forming pores [11,12]. These defects can lead to a decrease in mechanical properties and are a major obstacle to part certification. Due to the optical opacity of metal substrates, the in-process detection of subsurface phenomena by traditional optical and acoustic methods, such as the keyhole geometry, melt pool dynamics, and pore formation, is indirect and has certain limitations. Therefore, the in-situ diagnostics that can detect the subscale dynamics is crucial for the melt pool dynamics investigation and identify the defect formation mechanisms [13]. Till now, several researchers have used advanced X-ray diffraction technology and high-speed imaging to capture the dynamic behavior of the keyhole, revealing the relationship between process parameters, keyhole oscillation and collapse, as well as the pore formation [14–16]. For example, Cunningham et al. [17] observed the process of the laser-matter interaction by in-situ ultrahigh-speed synchrotron radiation imaging technology, and revealed the transition of the melt pool morphology from the conduction mode to the keyhole mode. They also summarized the keyhole generation mechanism into five stages. Zhao et al. [18] used the high-speed X-ray imaging technology to reveal the typical keyhole-induced pore formation process. The critical keyhole instability generates acoustic waves, which provides vital driving force for the pores near the keyhole tip to move away from the keyhole and become trapped as defects. Fan et al. [19] demonstrated that the crucial factor for keyhole instability was the flow vortex-induced protrusion on the rear keyhole wall. They also revealed that applying a transverse magnetic field could suppress the keyhole instability.

During the process of laser melting metals, the heat transfer in the melt pool is mainly through heat conduction and convection. For the low energy input cases (conduction mode), Marangoni effect is usually the dominant factor in the melt pool [20]. While in the deep keyhole state, recoil pressure has a significant effect on the deep flow of the melt pool and the keyhole morphology [21,22]. Strong fluid flow is also a major

feature of the high-power laser forming melt pool. Violent and complex fluid flow will not only lead to surface defects such as spattering and balling [23–25]. It also will influence the temperature distribution inside the melt pool and the mixing of alloy elements, which in turn affects the solidification structure and defect formation [26,27]. For example, Hoojatzadeh et al. [28] found that the high thermocapillary force, induced by the high temperature gradient in the laser interaction region, could rapidly eliminate pores from the melt pool during the LPBF process by combining in situ high-speed high-resolution synchrotron x-ray imaging experiments and multi-physics modeling. Yao et al. [27] revealed that when the melting mode changes from the conduction mode to the keyhole mode, the strong recoil pressure and Marangoni effect would enhance the fluid flow in the melt pool, resulting in interpenetrating interface morphology during the two dissimilar alloys manufacturing.

At present, the in-situ observation of laser melting metal process through synchrotron radiation technology mainly focuses on the relationship between the pore formation mechanisms and the keyhole characteristics, as well as the melt pool dynamics. While the effect of keyhole on the fluid flow in the melt pool is rarely reported. Besides, the bubble evolution, and the process of bubbles being captured by the solid/liquid (S/L) interface to form pores during the high-power additive manufacturing are also unclear.

Here, in order to deeply understand the basic physical phenomena during the high-power laser-matter interaction, the synchrotron radiation in-situ imaging technology is used to observe the process of laser melting aluminum alloy. By comparing and analyzing the metallurgical behaviors of the fluid in the melt pool before and after the laser is turned off under high energy heat input, the influence of the keyhole on the fluid flow is determined, the bubble generation mechanisms, evolution mechanisms and physical phenomena of the pore formation process are revealed. The metallurgical characteristics of the melt pool after the high-power laser is turned off are explored. Based on the basic research on the high-power laser manufacturing process, it is hoped that theoretical support can be provided for optimizing the process, reducing defects and improving the performance of parts.

2. Experiment

The high-speed and high-resolution X-ray imaging system at the BL16U2 beamline of the Shanghai Synchrotron Radiation Facility (SSRF) is used to study the metallurgical behaviors and bubble evolution in the melt pool during the high-power laser additive manufacturing in real time. The schematic of the in-situ X-ray imaging system is shown in Fig. 1. The transmitted ray beam is captured by the detection system and converted into visible light image, and then recorded by a high-speed camera (HAMAMATSU, 2048 × 2048 pixels, 1.3 μm spatial resolution, 16-bit depth) at 8000 fps, with the view field size of 2.6 mm × 1.7 mm. The high-power laser additive manufacturing system is equipped with the infrared fiber laser (1060 nm). The maximum power is 3000 W, and the laser spot diameter (D) is 0.65 mm. The material used in this experiment is Al 1060 with a particle size distribution of 15–53 μm. During the high-power laser additive manufacturing, the interaction of laser-matter will result in more complex melt pool dynamics than those under the general printing conditions. In order to ensure the maximum clarity of the area near the surface, so as to better observe the metallurgical behaviors and bubble evolution in the melt pool, the laser scans the Al bare plate. Although the condition without the powder bed also neglects the powder-induced effects, like spattering, particle entrainment, bubble generation etc., it can promote a better analysis of the characteristics of the metallurgical behaviors within the melt pool during high-power laser-matter interaction. The size of the substrate is 130 mm × 30 mm × 4 mm. The process parameters used in this work are shown in Table 1. Based on the definition of input energy

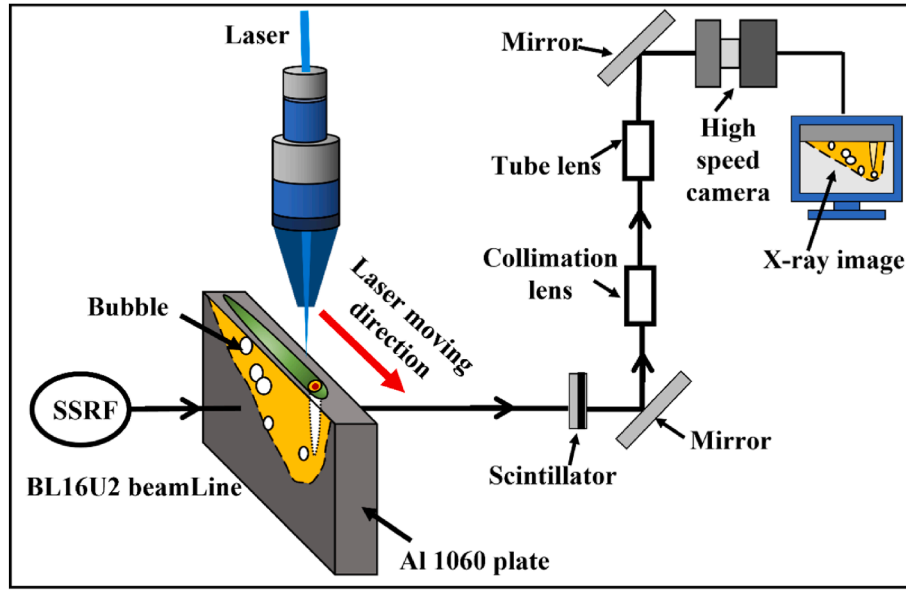


Fig. 1. Schematic of the in-situ X-ray imaging system.

Table 1

Process parameters used in this work.

Parameters	Laser power (P, W)	Scanning speed (V, mm·min ⁻¹)	Spot diameter (D, mm)	Energy density (E, GJ·m ⁻³)
Values	2700	3000	0.65	162.8

density E ($E = \frac{4P}{\pi D^2}$) [9], the energy density used in this work is about 162.8 GJ m^{-3} . Ar gas is used as the shielding gas. The X-ray imaging in laser additive manufacturing is based on the principle of phase

contrast method. The keyhole is filled with metal vapor, which has low X-ray absorptivity, appearing as bright area. The melt pool is filled with liquid, having higher absorptivity, appearing as grayscale area. The solid has the highest absorption, appearing as dark area. Therefore, the keyhole boundary is the transition between the bright and grayscale areas, and the melt pool boundary is the transition between gray and dark areas. The collected time-series images are analyzed by the imageJ software frame by frame to determine the fluid flow direction in the molten pool, the bubble generation mechanisms, the evolution mechanisms and the dynamic formation process of pores.

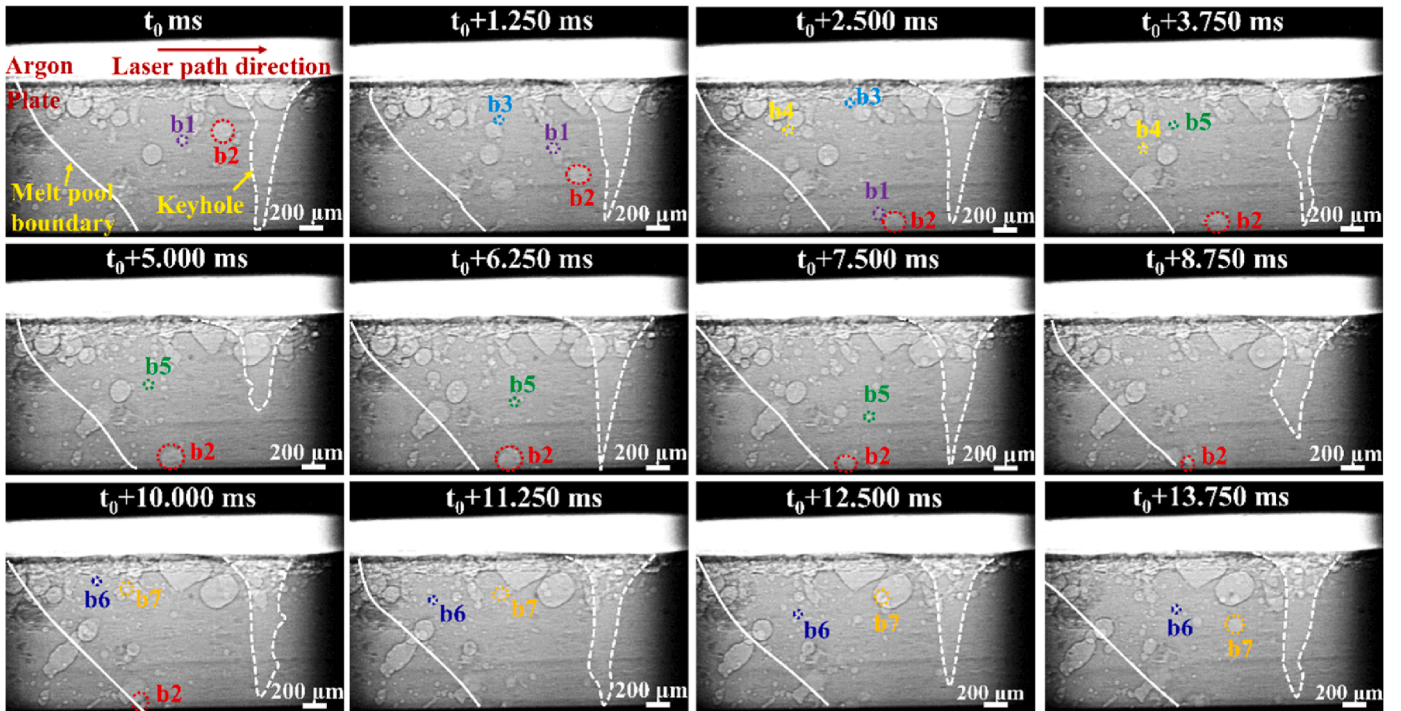


Fig. 2. Bubble positions under different moments in the melt pool. “ t_0 ” represents the reference time in the current figure. The bubbles b1, b2, b3, b4, b5, b6 and b7 are marked by the dotted circles with the color of purple, red, light blue, yellow, green, dark blue and orange, respectively. The complete process can be referred to the supplied video (named as video for fig. 2).

3. Results

3.1. Effect of keyhole on the flow

3.1.1. The flow characteristic with the keyhole

The flow direction in the melt pool can be determined by the bubble trajectory. The bubbles in the melt pool are marked under different moments, as shown in Fig. 2. In order to obtain the bubble trajectory, the melt pools under different moments are overlapped together. To clearly distinguish the time sequence, the transparency of the bubble color is set from high to low in the time sequence, that is, the earlier the time, the higher the transparency. Then the centers of each bubble are connected in sequence to form the moving path of the bubble in the time sequence, as shown in Fig. 3a. It is worth noting that due to the limitation of the view field size, the whole melt pool is not completely captured. Actually, according to the literatures, during the deep keyhole mode, the keyhole is close to the front wall of the melt pool, that is, the fluid in the front part of the melt pool (before the keyhole) is very thin [10,21]. The metallurgical behaviors of the fluid are mainly related to the keyhole and the melt pool in the back part. It can be seen from the figure that there are two main flow directions in the melt pool. One is the counterclockwise flow in the upper part of the melt pool (indicated by ①), and the other is the clockwise flow in the middle and lower part of the molten pool (indicated by ②), as shown in Fig. 3b.

3.1.2. The flow characteristic without the keyhole

In order to explore the effect of the keyhole on the fluid flow in the melt pool, the bubbles in the melt pool are also marked after the laser is turned off, as shown in Fig. 4. Fig. 4a shows the moment when the laser is just turned off and the keyhole is collapsing to pinch off a bubble at the bottom. Two bubbles are marked and their moving paths are determined according to the above method, as shown in Fig. 5a. It can be seen that after the laser is turned off, the flow direction in the molten pool is mainly counterclockwise (flow ①). Compared with the situation that the laser is on, the clockwise flow disappears.

During the laser-matter interaction process, the high-power laser leads to the formation of the keyhole, which is a deep and narrow vapor cavity formed by laser-induced metal vaporization. The presence of a keyhole significantly affects the temperature distribution, pressure distribution, and fluid behaviors within the melt pool. Under the irradiation of the laser, there is a significant temperature gradient on the surface of the melt pool, gradually decreasing from the center of the keyhole to the outside. For most metals (such as aluminum alloys), the surface tension decreases with increasing of the temperature, i.e., the surface tension gradient is negative [29]. According to the Marangoni effect, the liquid metal flows from the low surface tension area (high temperature, near the keyhole) to the high surface tension area (low temperature, edge of the melt pool). In the upper part of the melt pool (near the surface), the liquid metal flows outward from the center of the keyhole, then sinks and flows back along the edge of the molten pool,

forming a counterclockwise flow ①. The high temperature inside the keyhole causes the metal to vaporize, and the accompanied recoil pressure pushes the liquid metal downward, especially at the bottom of the keyhole. After the liquid metal is pushed to the bottom, it flows backward along the bottom and then flows upward to the keyhole, forming a clockwise flow ②. The existence of the deep keyhole increases the depth of the melt pool, forming two relatively independent flow loops.

After the laser is turned off, the energy input stops abruptly. The recoil pressure in the keyhole drops rapidly and disappears, and the keyhole collapses rapidly. The clockwise flow ② loses its main driving force, and the melt pool turns from a high-energy dynamic state to a cooling and solidification state. But the temperature gradient on the surface of the melt pool still exists (although it is weakening) because the center of the melt pool is still hotter than the edge. The Marangoni effect continues to drive the liquid metal to flow from the center to the edge, then sinks and flows back, forming a counterclockwise flow ①. Due to inertia, this flow pattern will continue for a while even if the laser is turned off until the molten pool is completely solidified.

3.2. Bubble formation mechanisms

In the previous section, the flow directions in the melt pool are analyzed based on the movement path of the bubbles. In this section, the bubble generation mechanisms observed in the in-situ experiment will be discussed.

Fig. 6 shows a bubble formation mechanism, where a bubble is formed by the entrainment of the shielding gas through the free surface. At t_0 moment, the gas starts to enter the melt pool, which is indicated by red circle. Then, the bubble migrates in a counterclockwise direction. At the moment of $t_0+1.500$ ms, the bubble contacts the melt pool boundary. At the time of $t_0+2.625$ ms, a pore is formed. The reason for the gas entering the melt pool can be attributed to that the violent flow leads to shear force on the free surface, disturbing the surface of the liquid metal, forming ripples or depressions, and entraining the shielding gas into the melt pool to form bubbles.

Fig. 7 shows the second bubble generation mechanism, in which bubbles are generated along the melt pool boundary. It can be seen from the figure that at $t_0+1.25$ ms, a small bubble forms at the melt pool boundary. At $t_0+2.50$ ms, the initially formed bubble becomes larger, and two more bubbles are formed. As the time increases, more bubbles appear at the melt pool boundary, and they expand with the time. When the bubble size is large enough, it will break away from the boundary and enter the melt pool. The melt pool boundary is the interface between the liquid and the solid, which has the largest temperature gradient, and is usually the region where solidification begins. As the solubility of gas in the solid is much lower than that in the liquid, when the S/L interface advances from the edge or bottom to the melt pool center, it is accompanied by a sharp drop in the solubility of the gas, and the gas dissolved in the melt precipitates and gathers at the boundary to form bubbles.

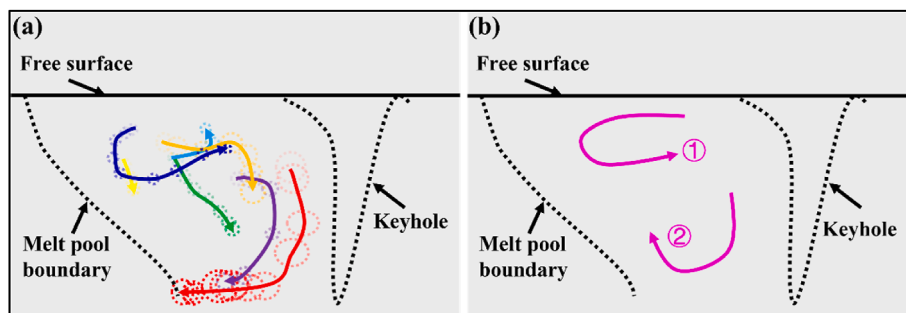


Fig. 3. Metallurgical behaviors in the melt pool, (a) schematic of the trajectory of the marked bubbles in Fig. 2, (b) schematic of the flow directions in the melt pool. The colors in (a) corresponding to the bubbles in Fig. 2. The flows ① and ② in (b) represent the counterclockwise flow and clockwise flow in the melt pool, respectively.

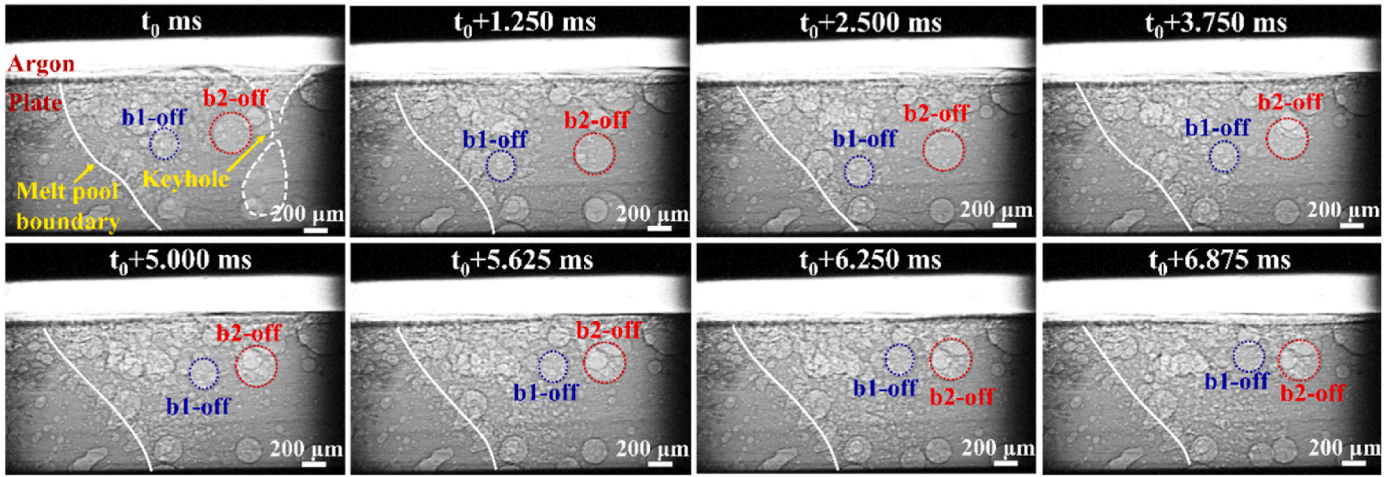


Fig. 4. Bubble positions under different moments in the melt pool after the laser is turned off. “ t_0 ” represents the reference time in the current figure. Different bubbles are marked by different colored circles. The complete process can be referred to the supplied video (named as video for fig. 4).

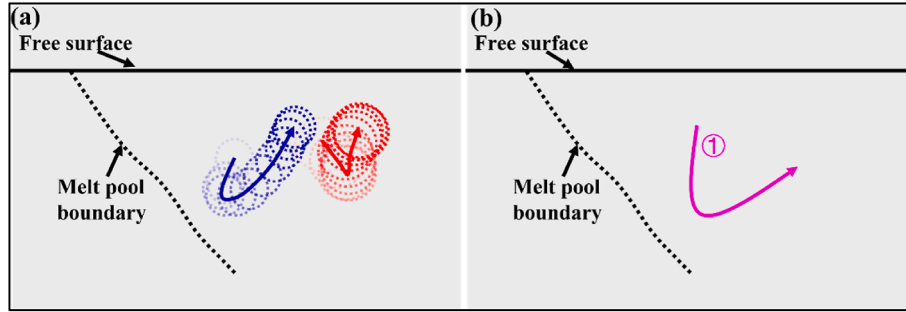


Fig. 5. Metallurgical behaviors in the melt pool after the laser is turned off, (a) schematic of the movement trajectory of the marked bubbles, (b) schematic of the flow directions in the melt pool.

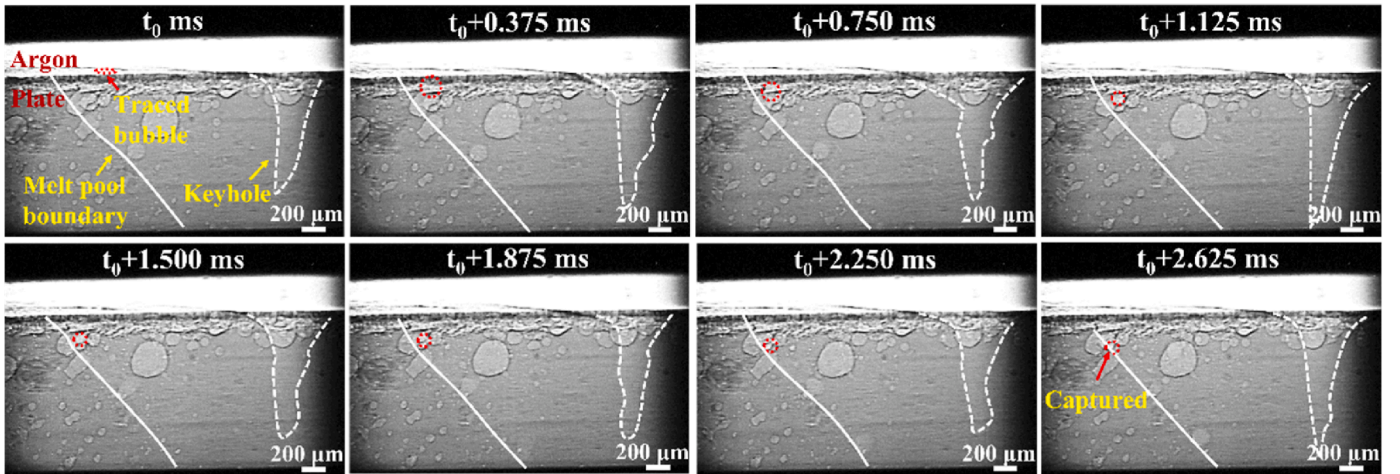


Fig. 6. Bubble generated by the entrainment of the shielding gas through the free surface. “ t_0 ” represents the reference time in the current figure. The bubble is indicated by the red circle. The complete process can be referred to the supplied video (named as video for fig. 6).

Fig. 8 shows the third bubble formation mechanism observed in this work, where a bubble is formed at the crack. The crack is indicated by blue arrows. It can be seen that a bubble generates along the crack at $t_0+0.375$ ms, and the volume of the bubble increased with time. At $t_0+2.875$ ms, the bubble detaches from the crack and enters the melt pool. As one of the defect forms, crack has high interface energy. In addition, the sharp edge of crack has high surface tension gradient. The

combined effects promote the aggregation of gas molecules to form a bubble. When the bubble volume reaches a certain level, it will detach from the crack and enter the melt pool under the action of buoyancy and flow disturbance.

The fourth bubble generation mechanism is to form a bubble at the keyhole bottom (Fig. 9). During the high-power laser additive manufacturing process, the temperature of the metal surface rises

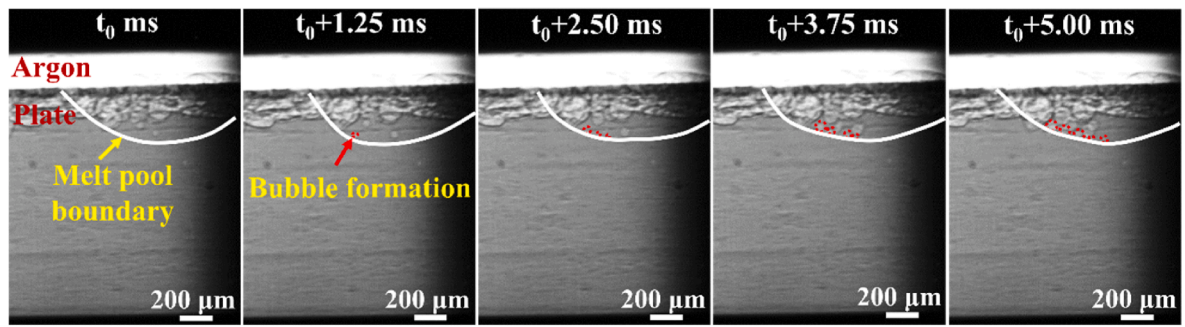


Fig. 7. Bubbles generated at the melt pool boundary. “ t_0 ” represents the reference time in the current figure. The bubbles are indicated by the red circle. The complete process can be referred to the supplied video (named as video for fig. 7).

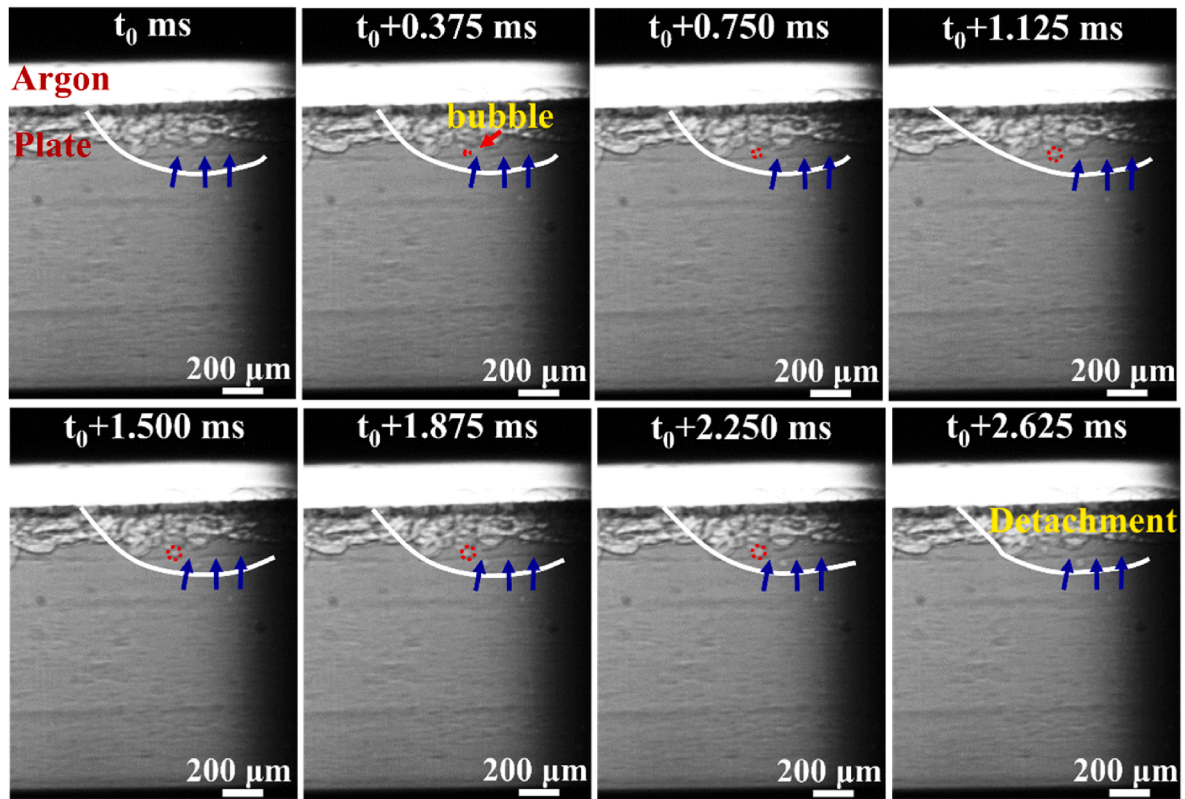


Fig. 8. Bubbles generated at the crack. “ t_0 ” represents the reference time in the current figure. The bubbles are indicated by the red circle. The crack is indicated by the blue arrows. The complete process can be referred to the supplied video (named as video for fig. 8).

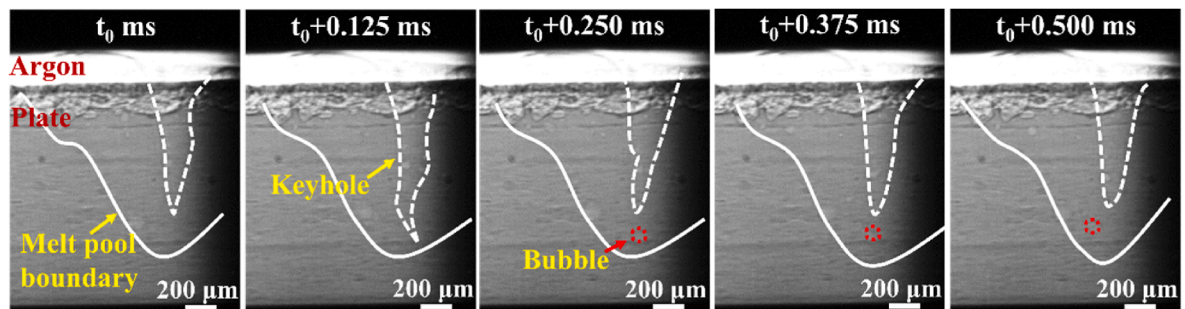


Fig. 9. Bubble generated by the keyhole fluctuation. “ t_0 ” represents the reference time in the current figure. The bubble is indicated by the red circle. The complete process can be referred to the supplied video (named as video for fig. 9).

rapidly and exceeds the boiling point. The recoil pressure brought by the metal vapor pushes the free surface away to form a keyhole. The keyhole bottom is directly exposed to the strong irradiation of the laser, resulting in a rapid increase in the keyhole depth. Theoretically, for given process parameters, due to the law of energy conservation, excluding the heat loss in the forms of heat conduction, radiation, convection, etc., the keyhole has a theoretical maximum depth. Based on the discussion in 3.1 section, the collision of counterclockwise ① and clockwise ② will lead to bulges generated on the keyhole wall, blocking the downward irradiation of the laser beam and causing disturbances to the keyhole. When a bridge is formed between the front and rear keyhole walls, a bubble will be separated from the bottom of the keyhole, accompanied by a rapid decrease in the keyhole depth. Compared with the other three bubble generation mechanisms, the bubble formed at the keyhole bottom has smaller chance of escaping from the free surface of the melt pool due to the larger escape distance.

Fig. 10 shows the schematic of the bubble generation mechanisms observed in this work. The symbols i, ii, iii, and iv represent the bubble generated by the gas entrainment through the free surface, bubbles formed at the melt pool boundary, the crack and the keyhole bottom, respectively. According to the results and discussions above, the bubble formation process at high energy locations such as the melt pool boundary and the crack includes the accumulation of gas to form bubbles, and then the bubbles detaching under the action of buoyancy and fluid convection. Therefore, these two types of mechanisms tend to generate small bubbles. In comparison, the bubbles formed by gas entrainment and at the bottom of the keyhole are large.

3.3. Bubble evolution mechanisms

After the analysis of the bubble formation mechanisms, in this section, we will discuss the bubble evolution mechanisms.

Fig. 11 shows the entire process of a bubble escaping from the surface of the melt pool. According to the position of the bubbles, it can be seen that the bubble moves in the counterclockwise direction and escapes from the upper surface of the melt pool at time of $t_0+1.50$ ms. The calculation shows that the speeds of the S/L interface moving forward and the bubble escaping are around $0.0076 \text{ m}\cdot\text{s}^{-1}$ and $1.30 \text{ m}\cdot\text{s}^{-1}$, respectively. This bubble evolution mechanism is of great significance for reducing porosity.

Fig. 12 displays the second bubble evolution mechanism. Two bubbles in the melt pool, which are named as b1-m and b2-m, merges into one bubble. At $t_0+0.250$ ms, bubble b2-m collides with bubble

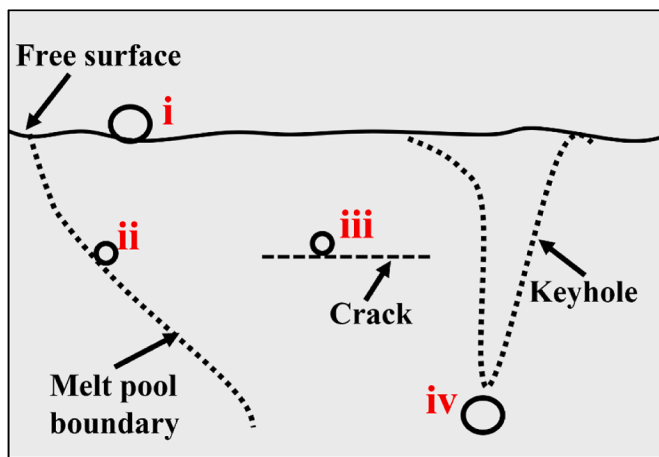


Fig. 10. The schematic of the different bubble generation mechanisms. The symbols i, ii, iii, and iv represent the bubble generated by the gas entrainment through the free surface, bubbles formed at the melt pool boundary, the crack and the keyhole bottom, respectively.

b1-m at a speed of $0.88 \text{ m}\cdot\text{s}^{-1}$. Then the large bubble (around $180 \mu\text{m}$) envelops the small bubble (around $100 \mu\text{m}$) at $t_0+0.375$ ms. At the moment of $t_0+0.500$ ms, two bubbles merge into one large bubble with a diameter of approximately $200 \mu\text{m}$, and a speed of $0.45 \text{ m}\cdot\text{s}^{-1}$.

Fig. 13 gives the third bubble evolution mechanism. At t_0 ms, the bubble with a diameter of about $190 \mu\text{m}$ moves towards the keyhole at a speed of $0.68 \text{ m}\cdot\text{s}^{-1}$. As the bubble is typically a closed area in the liquid filled with gaseous substance, which relies on surface tension to maintain its shape, after contacting with the keyhole ($t_0+0.125$ ms), the high temperature and high pressure surrounding the keyhole destroys the surface tension of the bubble, and the vapor in the bubble will be incorporated into the metal vapor. For high-power and low-speed additive manufacturing process, bubbles generated at the bottom of the keyhole are more likely to be captured by the keyhole. Because when the melt pool reaches dynamic equilibrium during the laser-matter interaction, the keyhole movement speed is almost equal to the laser scanning speed. Therefore, the keyhole moves slowly under low scanning speed. Compared with the long distance for escaping from the free surface of the melt pool, the keyhole is kind of giving the bubble a new escape interface.

Fig. 14 shows the dynamic process of a bubble being captured by the S/L interface to form pores. At the moment of t_0 ms, the bubble with a diameter of about $190 \mu\text{m}$ moves towards the melt pool boundary at a speed of $1.33 \text{ m}\cdot\text{s}^{-1}$, and then captured by the solidification interface at $t_0+0.75$ ms. During this process, the S/L interface moves with a speed of $0.009 \text{ m}\cdot\text{s}^{-1}$, much lower than that of the bubble. When the metal changes from liquid to solid, it will shrink in volume (increase in density), which will have a squeezing effect on the bubble, resulting in the volume shrinkage (indicated by the blue arrow). This extrusion may temporarily increase the internal pressure of the uncaptured part of the bubble, causing it to expand under the interface constraint. The advancement of the solidification interface may change the bubble shape from spherical to ellipsoidal or irregular to adapt to the geometric constraint of the S/L interface. At $t_0+3.00$ ms, the bubble splits into two parts from the solidification interface, in which one part forms a pore (I), and the other part forms a bubble (II) with a speed of $1.06 \text{ m}\cdot\text{s}^{-1}$ and a diameter of about $140 \mu\text{m}$. This split phenomenon may be related to the following factors. 1) Buoyancy effect. The uncaptured part of the bubble is subject to buoyancy in the melt pool. When the bubble volume increases to a certain size, and the buoyancy exceeds the adhesion or resistance of the solidification interface to it, the bubble will break away from the interface and enter the melt pool. 2) melt pool convection effect. In laser manufacturing, there is strong Marangoni convection driven by the surface tension gradient and thermal gradient in the melt pool. The flow provides additional shear force for the bubbles, helping them to break free from the constraint of the solidification interface. This new formed bubble II moves in the melt pool under the buoyancy and convection effect. At $t_0+4.50$ ms, it moves towards the solidification interface at a speed of $0.50 \text{ m}\cdot\text{s}^{-1}$. At the moment of $t_0+12.75$ ms, two pores are totally formed. Till now, this kind of pore formation process has not been reported yet.

Fig. 15 also gives another case of a bubble being captured by the solidification interface. At t_0 ms, the bubble with a diameter of about $70 \mu\text{m}$ moves towards the melt pool boundary at a speed of $0.12 \text{ m}\cdot\text{s}^{-1}$. From the moment that the bubble and the S/L interface come into contact, to the complete capture of the bubble to form a pore, no small bubble is separated from the original bubble, only the change in shape. Comparing the two cases (Figs. 14 and 15), it can be inferred that the reason for the different phenomenon may be related to the bubble size. For small-sized bubbles, there is not enough volume change to drive their deformation and separation. This kind of pore formation has also been reported in other literature [21]. Due to the unbalanced forces in the melt pool, the bubble shape can be irregular. Here, the commercial software ImageJ is used to determine the area of the bubbles, and then the average diameter is calculated to preliminarily determine the critical

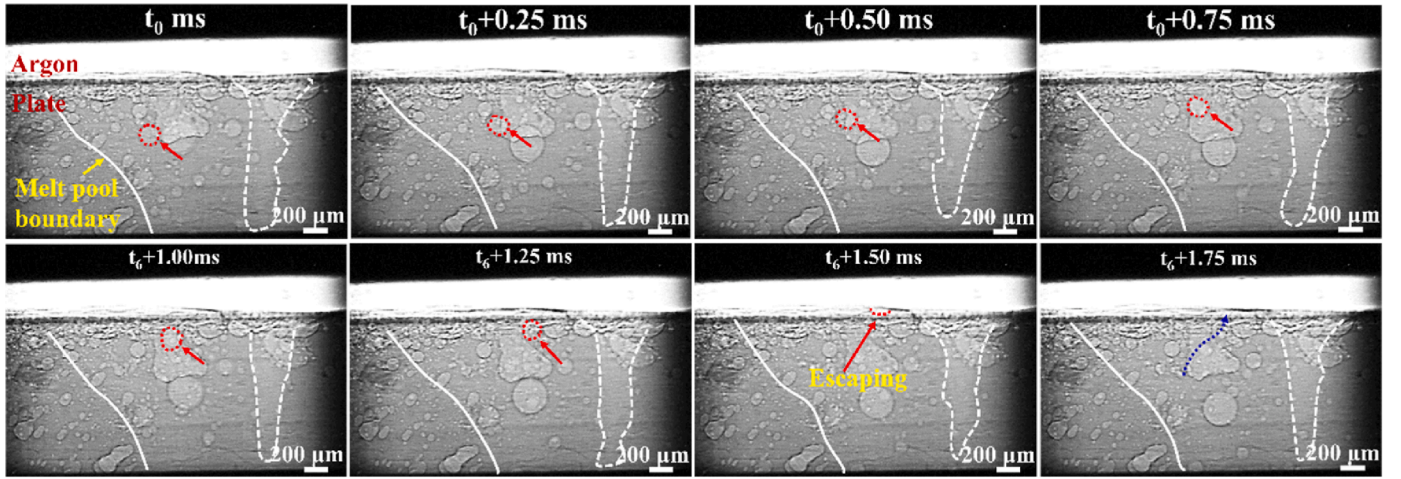


Fig. 11. Bubble escaped from the free surface of the melt pool. “ t_0 ” represents the reference time in the current figure. The bubble is indicated by the red circle. The blue arrow indicates the moving direction of bubble. The complete process can be referred to the supplied video (named as video for fig. 11).

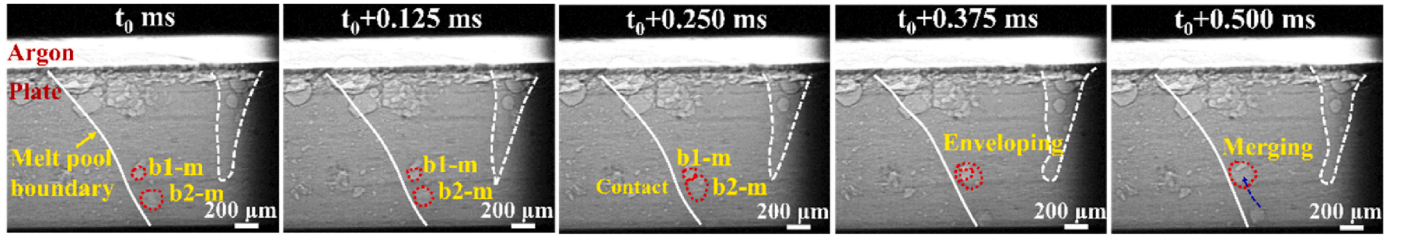


Fig. 12. Two bubbles merging into one big bubble. “ t_0 ” represents the reference time in the current figure. The bubbles are indicated by the red circles and named as b1-m and b2-m. The blue arrow indicates the moving direction of bubble b2-m. The complete process can be referred to the supplied video (named as video for fig. 12).

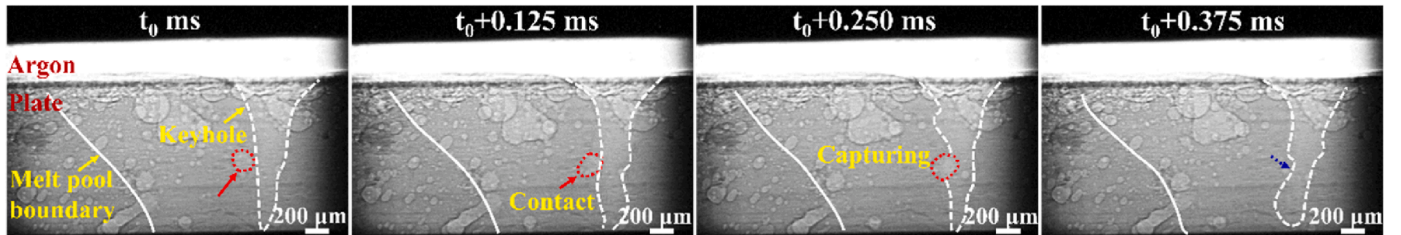


Fig. 13. Bubble disappearing via captured by the keyhole. “ t_0 ” represents the reference time in the current figure. The bubble is indicated by the red circle. The blue arrow indicates the moving direction of bubble. The complete process can be referred to the supplied video (named as video for fig. 13).

bubble size for splitting into small bubble during the capturing process by S/L interface. Table 2 shows the statistical results. Although the sample size is small, these bubbles are representative and statistically significant. This is because that during the statistical analysis, large bubbles that obviously will split, as well as too small bubbles that obviously will not split, are excluded to reduce the workload. In the last column of Table 2, the letters ‘Y’ and ‘N’ indicate there is split bubble, no split bubble, respectively. The results indicate that the bubble size around 70 μm can be considered as critical diameter.

The bubble evolution mechanisms observed in this work are demonstrated in the schematic form in Fig. 16 for better understanding and comparing.

3.4. Bubble characteristics with the laser off

Fig. 17 shows the characteristics of bubbles in the melt pool after the laser is turned off ($t_0+2.5$ ms). It can be seen that a large number of

bubbles appear after the laser is turned off, which may be attributed to the following physical processes. 1) Pressure fluctuations. In laser additive manufacturing, the huge recoil pressure is the driving force for the keyhole opening, while the surface tension is the driving force to close the keyhole. After the laser is turned off, the recoil pressure disappears immediately, accompanied by the breakdown of the dynamic equilibrium of the melt pool, resulting in strong pressure fluctuations and violent fluid flow in the melt pool. According to the definition of the Reynolds number ($Re=(\rho vL)/\mu$, in which ρ , v , L and μ indicate the fluid density, velocity, the characteristic length and viscosity, respectively), it can be concluded that the Reynolds number will be increased and the turbulence will be enhanced, causing the entrainment of the shielding gas. Besides, the turbulent flow will also induce the bubble breakup [30, 31]. 2) Solubility reduction. The gaseous solubility in the melt is not only related to the temperature, but also to the pressure. After the laser is turned off, the temperature drops sharply and the recoil pressure disappears immediately, which leads to a sharp drop of gaseous solubility,

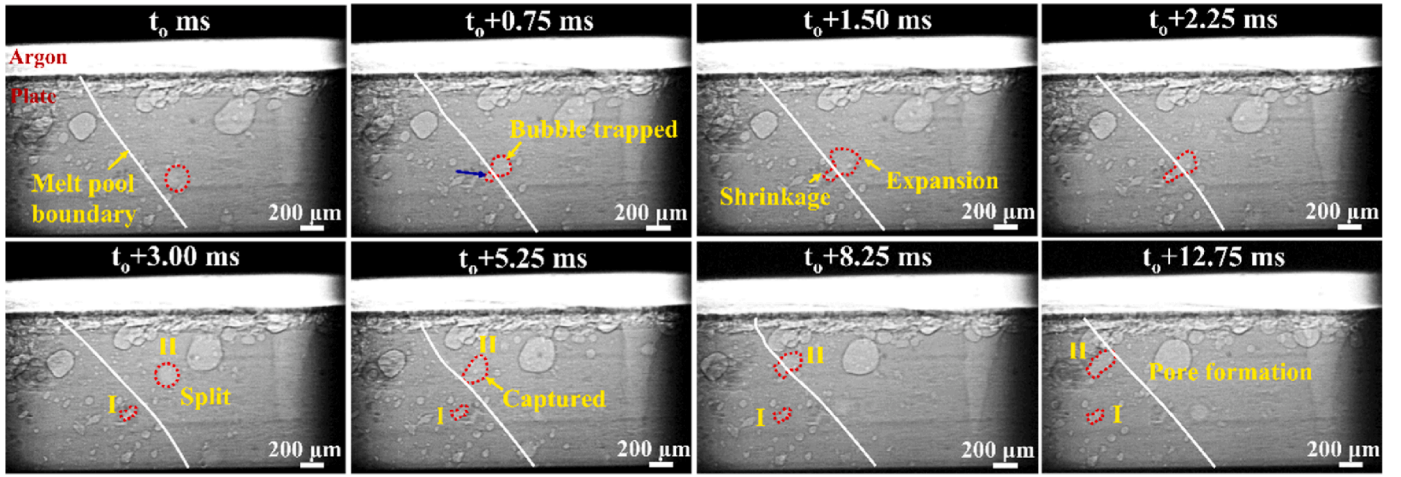


Fig. 14. The dynamic process of the bubble being captured by the S/L interface, in which a small bubble is split from the origin one and then also captured by the S/L interface. “ t_0 ” represents the reference time in the current figure. The bubbles are indicated by the red circle. The origin bubble is named as “I” and the split bubble is named as “II”. The blue arrow indicates the captured part of the bubble. The complete process can be referred to the supplied video (named as video for fig. 14).

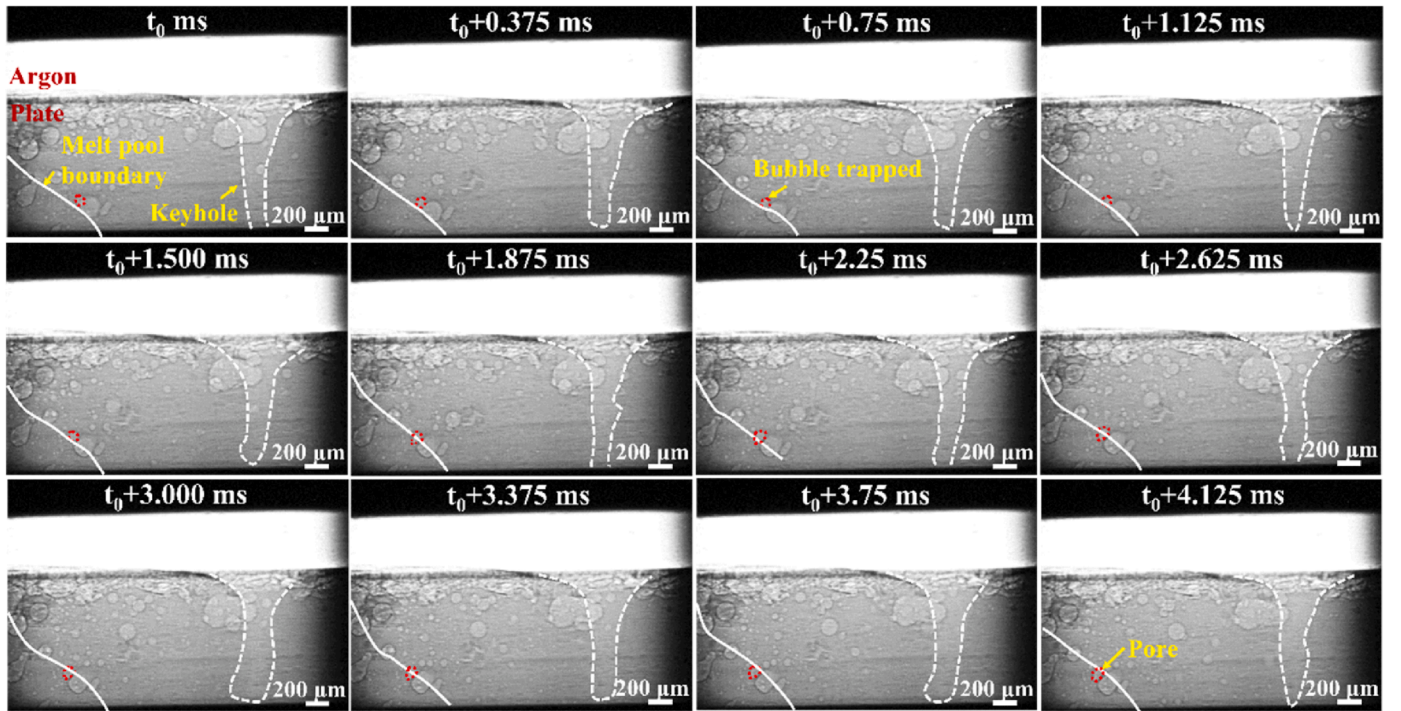


Fig. 15. A bubble being captured by the S/L interface, without a small bubble formed. “ t_0 ” represents the reference time in the current figure. The complete process can be referred to the supplied video (named as video for fig. 15).

thereby inducing the nucleation and growth of a large number of bubbles. 3) Rapid solidification constraint. When the melt pool begins to solidify, the liquid area shrinks rapidly, and the bubbles are compressed or squeezed, and may be forced to split into smaller units to adapt to the shrinking space. This corresponds to the bubble being captured by the solidification interface discussed above, in which new small bubbles may be formed under the interface constraint. The moment of $t_0+37.5$ ms shows the final solidified state, in which a large number of pores are formed. This is different from the phenomenon reported in the literature for low-power additive manufacturing, in which only a large vapor bubble is formed due to the keyhole collapsing [32,33]. The complete process of the bubble evolution in the melt pool after the laser is turned off can be referred to the supplied video (named as video for Fig. 17).

4. Conclusion

In this work, the synchrotron radiation technology is used to in situ observe the fluid flow, generation and evolution of the bubble, and dynamic process of bubbles being captured by the solid/liquid interface in the melt pool during high-power additive manufacturing. The main conclusions are as follows.

1. The occurrence of deep keyhole increases the melt pool depth, leading to two relatively independent flow loops. One is the counterclockwise flow in the upper part of the melt pool, and the other is the clockwise flow in the middle and lower part. While there is just one counterclockwise flow direction in the melt pool without the

Table 2

Bubble size and whether there is splitting during the capture process.

No.	Area (μm^2)	Average diameter (μm)	Splitting small bubble
1	25277	179.40	Y
2	24722	177.42	Y
3	21284	164.62	Y
4	7291	96.35	Y
5	4444	75.22	Y
6	3910	70.56	N
7	3750	69.10	N
8	3720	68.82	N
9	2378	55.03	N
10	1944	49.75	N
11	1388	42.04	N
12	1371	41.78	N
13	1215	39.33	N

Letter 'Y' indicates there is split bubble, and 'N' indicates no split bubble.

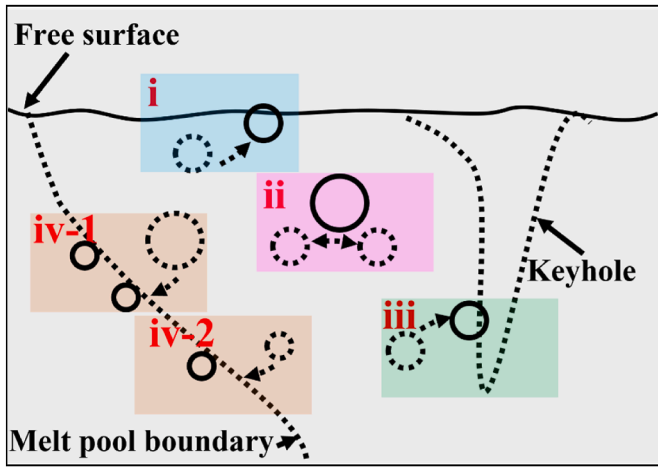


Fig. 16. Schematic of the bubble evolution mechanisms. The symbols i, ii, iii, and iv represent the evolution mechanisms of escaping from the free surface, emerging into a large bubble, capturing by the keyhole, and capturing by the solidification interface, respectively. For the iv-1, it means that the large bubble splits into two pores, while for iv-2, only a pore is formed for the small bubble. The dotted circles indicate the bubble position for the previous time. The dotted arrow indicate the bubble moving direction.

keyhole. The main reason for the difference is that the recoil pressure caused by vaporization exerts a downward force on the fluid.

- Four kinds of bubble generation mechanisms are observed in the melt pool, namely, the bubble formed by the gas entrainment through the free surface, the bubble formed at the melt pool boundary, the bubble formed at the crack, and the bubble formed at the bottom of the keyhole.
- There are four main bubble evolution mechanisms observed in this work, namely, bubbles escaping from the free surface of the melt pool, bubbles merging into a large bubble, bubbles captured by the keyhole, and bubbles captured by the solidification interface to form pores.
- The volume and morphology of bubbles will change after being captured by the S/L interface. During the capture process, under the interface constraint, the large-sized bubbles splits into a pore and a new small bubble. That is, large-sized bubbles lead to more pores. While the small-sized bubble completely forms a pore.
- After the laser is turned off, a large number of small bubbles occur in the melt pool and finally turn into pores, which may be related to the following factors: pressure fluctuations, solubility reduction, and the rapid solidification constraint.

This work shows the phenomena in high-power additive manufacturing, which are different from those in low-power additive manufacturing. Two independent circulation flows occur in the obtained large-sized melt pool, and the large-sized bubbles split into small-sized pores during the capture process, as well as a large number of small bubbles occurs in the melt pool after the laser is turned off. The results can provide theoretical support for optimizing processes and reducing defects for high-power additive manufacturing, as well as the welding process. For example, during the actual printing, reducing the turbulence can be considered to minimize pore formation, such as applying an external magnetic field or adjusting the scanning strategy. Furthermore, a stable flow field can also help to promote bubbles to escape.

CRediT authorship contribution statement

Liping Guo: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hanjie Liu:** Writing – review & editing, Data curation. **Zijue Tang:** Writing – review & editing, Methodology. **Yuhan Qian:** Investigation. **Zhenyang Gao:** Writing – review & editing. **Yi Wu:** Writing – review & editing, Supervision. **Haowei Wang:** Supervision. **Pei Wang:** Writing – review &

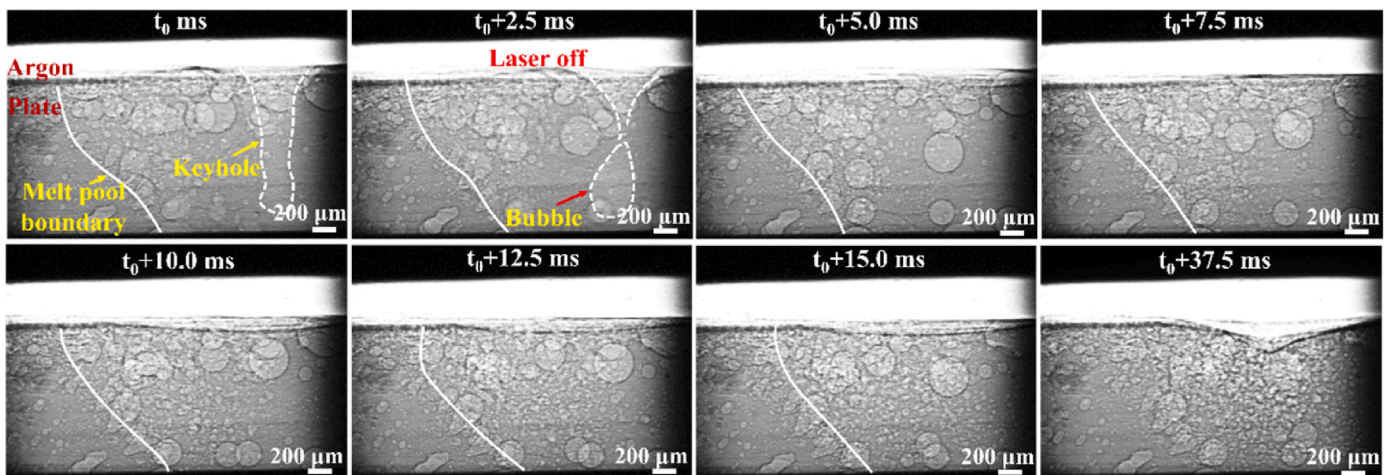


Fig. 17. The bubble characteristics in the melt pool after the laser is turned off. “ t_0 ” represents the reference time in the current figure. The complete process can be referred to the supplied video (named as video for fig. 17).

editing. **Xingxing Wang**: Writing – review & editing. **Yongbing Li**: Writing – review & editing. **Hongze Wang**: Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.tramat.2025.100084>.

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